

ECON 407 Behavioral Economics

Fall 2026

Instructor: Mu Zhang (muzhang@umich.edu)

GSI: Jackson Rockmael (rockmael@umich.edu)

Logistics

Prerequisite: Enforced: minimum grade of C– in ECON 401 or graduate standing. Advisory: STATS 250, or ECON 249, or STATS 426, or ECON 451.

Classes: Two lectures per week on **Monday and Wednesday from 10:00 a.m. to 11:20 a.m. at 2255 North Quad.**

Website: Canvas. This syllabus, homework, lecture notes or slides, and other relevant documents will be made available through the course website.

Office hours of the instructor: **Mondays from 4:00 p.m. to 5:00 p.m. at 4417 North Quad.**
Additional office hours will be provided before the midterm and the final exams.

Office hours of the GSI: **Thursdays from 4:00 p.m. to 5:00 p.m. It will be on Zoom**
<https://umich.zoom.us/j/9396262769> **before a room is assigned.**

Please check the Canvas for the most updated syllabus during the semester. I will also post an announcement if there is a major change.

Course Description

Behavioral economics studies how people make choices. It is an active field drawing together ideas and insights from economics, psychology, political science, and management, among other fields. Standard economic theory is based on the idea that economic agents are rational. Empirical research, however, often discovers discrepancies between rational behavior and people's actual behavior. Behavioral economics questions the assumption of perfect rationality and uses those discrepancies to improve both theoretical and empirical aspects of standard economic theory. The main objective of the course is to introduce students to this field.

To accomplish its goals, behavioral economics does not start from scratch, but typically builds on standard economic theory by changing some assumptions of standard economic models to obtain more realism. Behavioral economics uses insights from psychology to decide which assumptions need to be revised and how to best revise them to make the models more realistic. Our course will follow this tradition. We will first introduce standard economic theory, point out its shortcomings, and then introduce behavioral models that incorporate insights from psychology.

Below is a tentative outline of the course. Roughly speaking, there are two types of decision problems: static ones and intertemporal/dynamic ones. Our course starts with an investigation of **static choice without uncertainty**, in which we study empirical phenomena such as endowment effect,

status-quo bias, etc. Then, we study **static choice under uncertainty**. The standard model of decision-making under uncertainty is the expected utility model. We describe experimental work from both economics and psychology, such as the Allais (1953) paradox, to illustrate violations of the expected utility model. Then, several models that explain these violations will be introduced. The most influential one is prospect theory of Kahneman and Tversky (1979). This model introduces reference dependence and loss aversion to the expected utility model. We will also talk about the difference between risk and uncertainty/ambiguity, the Ellsberg (1961) paradoxes, and the maxmin expected utility model of Gilboa and Schmeidler (1989) that can accommodate aversion to ambiguity.

Next, we study **intertemporal choice**. The standard model of intertemporal choice in economics is the exponential discounting model. After introducing this model, we again go on to describe experimental violations of it. To account for these violations, we introduce and analyze the quasi-hyperbolic discounting model, time inconsistency (i.e., a decision maker's current self and future selves disagree on which dynamic choices are optimal), and sophistication (i.e., a decision maker's current self understands her future selves' preferences and how they differ from her own) and naïveté (i.e., a decision maker's current self does not understand her future selves' preferences and how they differ from her own).

Last, if we have time (likely we won't), we will talk about **social preferences**. Standard economic theory assumes that people do not care about the process that leads to a particular consequence, and they care only about their own payoffs. Many experiments, however, illustrate that people care about both the process that leads to a consequence and the perceived fairness of it. To study these issues, we need to have a theory of how agents interact with each other. The standard tool in economic theory to study strategic interactions is game theory. We introduce game theory briefly and experiments that question standard assumptions of certain games. Then, we introduce social preferences to game theory to accommodate experimental findings.

Email Policy

If something about lectures, readings, or homework is not clear, the most efficient way is to come to **my or the GSI's office hour**.

If you would like to ask questions via email, **please first send it to the GSI**. You can contact me if you do not hear back from him within 24 hours.

Problem Sets, Class participation, Exams, and Grades

Your grade for the course is based upon your weighted performance in several areas. These areas and the corresponding weights are indicated below in tabular form.

Task	Percentage of grade
Problem sets	15
Class participation	10
Midterm	35
Final	40

Problem sets:

There will be up to six problem sets. They will be **graded based on effort**. The due dates will be announced when they are posted. The last problem set may not have a due date. You are encouraged to work in groups, but each of you must write down and submit your own homework. Please submit your homework (**ideally scanned or typed; a compiled PDF file of clear photos also works**) on time to Canvas Assignments. **Late submissions without a legitimate reason approved by the instructor in advance will not be graded.**

The main purpose of problems sets is to familiarize you with applying the behavioral models learnt in class and prepare you for the closed-book exams. You are encouraged to use AI to learn course materials, but please **Do Not** use it for generating homework solutions. Afterall, they are not graded based on correctness.

The Economics Department Policy on Graded Assignments can be found here:

http://lsa.umich.edu/content/dam/econ-assets/Econdocs/Graded%20Assignments_updated%206_2016.pdf

Class participation:

You are expected and strongly encouraged to attend the lectures in person. There will be **6 or 7** “surprise” (i.e., without notice in advance) quizzes in class. **Missing up to two of them without excuses will not affect your grade for class participation.** For excused absences, please refer to the information below.

There is nothing to prepare for those quizzes. They often take the form of an **in-class experiment (we are studying behavioral economics together)** and there are no right or wrong answers. Simply finishing a quiz guarantees full grade for it.

Lecture recordings:

Lectures are not recorded automatically. If you have an approved excused absence, please email me and the GSI **before the lecture** to request that it be recorded.

Exams:

The midterm exam on **Wednesday, October 21** will be **closed-book** and **in-class (80 minutes)**. The final exam will be **non-cumulative, closed-book**, and its date and time (**Thursday, December 17, 10:30 am - 12:30 pm**) information can be found here:

<https://ro.umich.edu/calendars/final-exams/fall-2026>

If you have any known or anticipated conflict with the assignment or exam dates, and you have one of the acceptable excuses listed below, please let me and the GSI know as early as possible. Please make every effort to attend exams and submit the homework on time.

Excused absences generally consist of the following: illness of the student or a member of the student's immediate family; death of a member of the student's immediate family; university

sanctioned trips and activities; and major religious holidays. In each case, written verification will be required and permission to miss an exam must be secured before the scheduled exam time.

Members of athletic teams must present to the instructor, prior to each absence because of the membership on athletic teams, a written statement signed by the appropriate authority specifying the exact date of any such proposed absence.

Grades:

There may be a curve for the final letter grade. If there is a curve, it will be based on the total score. Individual exams are not subject to a curve. Otherwise, the course follows the Economics Department Policy on Graded Assignments in circumstances not explicitly spelled out in this syllabus, which can be found here:

http://lsa.umich.edu/content/dam/econ-assets/Econdocs/Graded%20Assignments_updated%206_2016.pdf

Academic Integrity

The course complies with the Economics Department policy on academic integrity, as posted on <http://lsa.umich.edu/content/dam/econ-assets/Econdocs/Academic%20Misconduct.pdf>.

Student Mental Health and Wellbeing

University of Michigan is committed to advancing the mental health and wellbeing of its students. If you or someone you know is feeling overwhelmed, depressed, and/or in need of support, services are available. For help, contact Counseling and Psychological Services at (734) 764-8312 and <https://caps.umich.edu/> during and after hours, on weekends and holidays, or through its counselors physically located in schools on both North and Central Campus. You may also consult University Health Service at (734) 764-8320 and <https://www.uhs.umich.edu/mentalhealthsvcs>, or for alcohol or drug concerns, see www.uhs.umich.edu/aodresources. For a listing of other mental health resources available on and off campus, visit: <http://umich.edu/~mhealth/>.

Books

There is **no textbook** for the course. A few recommended but not required books are below.

- Advances in Behavioral Economics (2004) by Colin F. Camerer, George Loewenstein, and Matthew Rabin
- Rational Choice in an Uncertain World: The Psychology of Judgment and Decision Making (2001) by Reid Hastie and Robyn M. Dawes
- Psychology of Judgment and Decision Making (1993) by Scott Plous
- Behavioral Game Theory (2003) by Colin F. Camerer
- Choices, Values, and Frames (2000) by Daniel Kahneman and Amos Tversky
- Handbook of Experimental Economics (1997) by John Kagel and Al Roth
- Behavioral Law and Economics (2000) by Cass Sunstein
- Advances in Behavioral Finance (2005) by Richard Thaler

Tentative Course Schedule

Our class time will include lectures, discussions, and in-class experiments. There may be changes to this outline.

Part 0

INTRODUCTION

Behavioral Economics: Past, Present, Future by Colin F. Camerer and George Loewenstein

Part 1

REFERENCE-DEPENDENT PREFERENCES

- Endowment Effect
- Status Quo Bias
- Loss Aversion

Loss Aversion in Riskless Choice: A Reference-dependent Model by Tversky and Kahneman

CONTEXT EFFECTS

- Similarity Effect
- Attraction Effect
- Compromise Effect

Part 2

PREFERENCES OVER RISKY AND UNCERTAIN OUTCOMES

- Risk Aversion: St. Petersburg Paradox
- Certainty Effect: Allais Paradox
- Risk Seeking Toward Losses
- Prospect Theory
- Risk versus Uncertainty
- Disappointment Aversion

Choices, Values and Frames by Daniel Kahneman and Amos Tversky

Prospect Theory in the Wild: Evidence from the Field by Colin F. Camerer

Part 3

AMBIGUITY

Are Risk Premium Anomalies Caused by Ambiguity? by Robert A Olsen and George H Troughton.

Part 4

INTERTEMPORAL CHOICE

- Samuelson's Exponential-Discounting Model
- Self-Control Problems and Hyperbolic Discounting
- Harmful Substances and Government Policy
- Anxiety, Optimism and Other Anticipatory Emotions
- Market Implications

Time Discounting and Time Preference: A Critical Review by Shane Frederick, George Loewenstein, and Ted O'Donoghue

Doing It Now or Later by Ted O'Donoghue and Matthew Rabin

Part 5 (If we have time...)

FAIRNESS AND SOCIAL PREFERENCES

- Selfishness
- Altruism
- Reciprocity and Intentionality

Fairness as a Constraint on Profit Seeking: Entitlements in the Market by Daniel Kahneman, Jack L. Knetsch, and Richard H. Thaler

A Theory of Fairness, Competition, and Cooperation by Ernst Fehr and Klaus M. Schmidt

Explaining Bargaining Impasse: The Role of Self-Serving Biases by Linda Babcock and George Loewenstein